

A Deep Learning-Based Intrusion Detection System for Safeguarding ROS 2-Powered Unmanned Ground Vehicles Against DoS Attacks

1st Eduardo Fraga da Silva
AI and Cyber Futures Institute
Charles Sturt University
Canberra, ACT, Australia
efragadasilva@csu.edu.au

2nd Fendy Santoso
AI and Cyber Futures Institute
Charles Sturt University
Canberra, ACT, Australia
fendy.santoso@gmail.com

3rd Lihong Zheng
School of Computing, Mathematics,
and Engineering
Charles Sturt University
Wagga Wagga, NSW, Australia
lzheng@csu.edu.au

Abstract—Recent advances in robotics have increased the reliance on networked middleware, such as Robot Operating System 2 (ROS 2), for orchestrating Unmanned Ground Vehicles (UGVs) in dynamic and safety-critical settings. However, this distributed architecture exposes platforms like the Turtlebot4 to Denial-of-Service (DoS) vulnerabilities, which can incapacitate critical sensor and control messages. This paper presents a deep learning-based framework for near-real-time detection of DoS attacks in a ROS 2-enabled UGV. A novel adversarial script, generated using a Large Language Model (LLM), systematically floods targeted ROS 2 topics to simulate genuine network-based intrusions. The resulting dataset, comprising both benign and adversarial conditions, undergoes rigorous cleaning, feature engineering, and one-second binning to capture key metrics such as burstiness and topic entropy. Four neural architectures are evaluated: a Multilayer Perceptron (MLP), a Long Short-Term Memory (LSTM), a 1D Convolutional Neural Network (CNN), and an unsupervised Variational Autoencoder (VAE). Experimental findings show that the MLP, leveraging the engineered features, achieves 97% accuracy and an AUC of 0.998 in classifying DoS traffic, outperforming sequence-based models and the VAE. Integrated Gradients analysis further reveals that metrics reflecting abrupt message spikes and imbalanced topic usage are highly discriminative of malicious behavior. These results underscore the impact of domain-specific feature design in safeguarding ROS 2-based robotic platforms against emerging cyber threats.

Index Terms—Deep Learning, DoS, Unmanned Ground Vehicle, and Robot Operating System (ROS).

I. INTRODUCTION

Unmanned Ground Vehicles (UGVs) have rapidly expanded their role in industrial and research applications, from automated warehouse operations [1] to hazardous exploration in search-and-rescue missions [2]. The Robot Operating System 2 (ROS 2) has become the standard middleware for developing such platforms, valued for its modular design and extensive open-source ecosystem [3]. However, ROS 2's distributed nature—where multiple nodes communicate via networked topics—opens vulnerabilities to cyberattacks. In particular, Denial-of-Service (DoS) attacks, whether targeting network protocols or exploiting vulnerabilities in the ROS 2 stack itself, threaten to disrupt critical sensor, control, and navigation messages by overwhelming the system with false

data [4]. In high-stakes, real-time environments, even brief disruptions can trigger cascading failures—delayed hazard detection, stalled navigation, or erratic robot behavior—potentially endangering both human operators and essential infrastructure. Moreover, transient network overloads can quickly escalate into sustained system-wide degradations, underscoring how even short-lived DoS attacks can have far-reaching operational and safety consequences. As the deployment of UGVs accelerates in mission-critical domains, ensuring rapid detection and mitigation of DoS attacks has become a pressing priority, vital to maintaining operational integrity and safety.

Deep learning approaches for network security have evolved significantly, from early applications of feed-forward networks for packet-level analysis [5] to sophisticated temporal models for sequence-based attack detection [6, 7]. Recent work has demonstrated the effectiveness of self-attention mechanisms in capturing complex network traffic patterns and improving attack detection accuracy [8], and graph neural networks [9] in capturing complex network interactions. In the context of robotics security, while proactive anomaly detection using multi-sensor fusion has shown promise in identifying unsafe physical robot behaviors [10], existing security solutions primarily rely on traditional anomaly detection approaches [11]. Recent quantitative frameworks have emerged that bridge this gap between traditional and learning-based approaches. Notably, Petho et al. [12] demonstrated how machine learning techniques can model complex interactions between attack parameters and system safety in vehicular networks, establishing a methodology for quantifying security risks that has implications for robotic systems. However, these approaches, while promising, have yet to fully address the unique challenges of ROS 2-based platforms, particularly in handling the complex temporal dependencies and heterogeneous data streams characteristic of robotic systems. These methods, though effective in certain environments, often struggle with the dynamic nature of real-world robotic deployments, particularly in handling multi-modal sensor streams, varying operational conditions, and cyberattacks like DoS that can disrupt critical ROS 2 communications.

The emergence of Large Language Models (LLMs) has

fundamentally changed the security landscape, offering adversaries sophisticated tools for automatically generating malicious code [13]. This capability enables higher-volume and more adaptive attacks than previously possible, challenging traditional detection methods. While deep learning has shown promise in detecting anomalies and intrusions in networked systems, existing approaches often overlook the unique challenges of ROS 2-based platforms: (1) the need to process multiple, heterogeneous sensor streams in real-time, (2) the complex temporal dependencies in robot control messages, and (3) the requirement for interpretable detection decisions in safety-critical operations. Motivated by these challenges, this work investigates: *How can we achieve near real-time detection of DoS attacks in physical ROS 2-enabled UGV deployments through interpretable deep learning that leverages ROS 2-specific traffic patterns, while maintaining high accuracy and low latency for safety-critical operations?*

To address this question, this paper introduces a novel DoS simulation framework that employs an LLM-generated flooding script targeting ROS 2 topics in a Turtlebot4 platform. Through data collection under both benign and adversarial conditions, the approach captures raw sensor and control messages, as well as detailed network statistics. These data go through an extensive feature engineering phase, paying particular attention to metrics such as *burstiness* (reflecting abrupt spikes in traffic) and *topic entropy* (capturing skewed or monopolized bandwidth usage). The study evaluates multiple neural architectures: a Multilayer Perceptron (MLP) [14] focused on tabular features, a Long Short-Term Memory (LSTM) [15] and a 1D Convolutional Neural Network (CNN) [16] designed to capture time-varying patterns, and a Variational Autoencoder (VAE) [17] operating under an unsupervised anomaly detection paradigm. Integrated Gradients (IG) [18] further clarify model decisions, highlighting the features or time windows most indicative of malicious behavior.

This work makes five key contributions: (1) a novel DoS attack simulation framework leveraging LLM-generated attack scripts, demonstrating vulnerabilities in current robotic systems, (2) an automatic comprehensive data cleaning and feature engineering pipeline introducing specialized metrics like burstiness and topic entropy, (3) comparative evaluation of multiple deep neural architectures tailored for ROS 2-based detection, providing insights into the effectiveness of different learning approaches, (4) extensive empirical analysis demonstrating high detection accuracy while identifying key factors in successful detection, and (5) interpretability analysis using IG to explain model decisions in the context of robotic security. The remainder of this paper is organized as follows: Sections III, IV, V, VII, and VI collectively detail the methodology, including data collection, feature engineering, and the architecture of the machine learning models. Section VIII presents the experimental results and analysis. Finally, Section IX concludes with implications for future research and practical applications in securing autonomous systems.

Experimental evaluations indicate that careful feature design and deep learning can achieve high-performance DoS detection in controlled ROS 2 settings. The MLP model achieves the highest overall performance, although the sequence-based

approaches emphasize the utility of short-term traffic patterns. Interpretability analysis shows that key ROS 2 topics (e.g., `/odom` or `/scan`) and dynamic measures (burstiness, topic entropy) consistently factor into successful detections, providing insights into how DoS attacks manifest and how best to mitigate them.

II. RESEARCH CONTRIBUTIONS

This paper presents a comprehensive framework for detecting DoS attacks in a ROS 2-based robotic platform, specifically focusing on the Turtlebot4 UGV. The primary goal is to address security vulnerabilities in modern robotic systems by leveraging deep learning models and interpretable Artificial Intelligence (AI) techniques. The key contributions are summarized as follows:

1) **Novel DoS Attack Simulation and Data Collection.**

A configurable DoS attack scenario is introduced on a physical ROS 2-enabled Turtlebot4 platform (Sections IV and III). Unlike previous works that rely on synthetic data generation [19], observer-based attack detection [20], or feature-based ML approaches with synthetic datasets [21], this research captures genuine attack behaviors and system responses through real-time data collection on an operational UGV, with millisecond-precision temporal resolution. This simulation is facilitated by AI-assisted malicious code generation using Anthropic Claude [22], illustrating how LLMs can be leveraged as instruments for sophisticated cyberattacks. Sensor and network data are comprehensively collected in both benign and adversarial conditions, covering multiple sensor modalities, real-time constraints, and 33 ROS topics. This yields a unique 3.8 GB dataset capturing the complexity of modern robotic traffic under DoS attacks, with high-fidelity temporal dynamics that synthetic and model-based approaches cannot replicate.

2) **Rigorous Data Cleaning and Feature Engineering.**

A multi-stage data processing pipeline (Section VI-A) is developed to remove artificially created ROS topics (specifically used to funnel DoS traffic), aggregate messages into 1-second bins, and perform feature scaling. By filtering out these inflated topics, the pipeline ensures training data remains representative of realistic traffic patterns, enhancing the model's generalization and improving detection across the ROS network. To capture complex attack dynamics, two specialized metrics are introduced: (a) *burstiness*, which highlights abrupt spikes in message volume, and (b) *topic entropy*, which exposes skewed or monopolized bandwidth usage (Section VI-D).

3) **Design and Evaluation of Deep Neural Architectures.**

Four deep learning architectures are implemented (Section VII):

- a) MLP for processing structured, tabular features,
- b) LSTM for capturing short-range temporal dependencies,
- c) 1D CNN for identifying local patterns and bursts,
- d) VAE for unsupervised anomaly detection when labeled data are scarce.

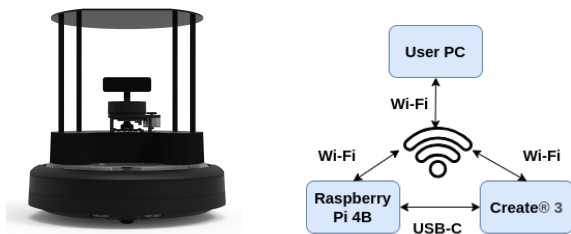
This diversity of models provides broad coverage of both time-series and latent-feature perspectives for DoS detection.

- 4) **Extensive Empirical Analysis and Benchmarking.** Each model is thoroughly evaluated (Section VIII) using accuracy, precision, recall, F1-score, and AUC. The MLP achieves excellent performance (97% accuracy, AUC $\approx$ 0.999), surpassing the sequence-based and unsupervised models. The results highlight the importance of feature design and model architecture.
- 5) **Integrated Gradients for Explainability and Trustworthiness.** Integrated Gradients analysis is applied to interpret each model’s predictions (Section VIII-B). This technique identifies critical input features (e.g., *burstiness* and *topic entropy*) that drive either classification or reconstruction. Such explainability is essential for building trust in AI-powered security pipelines and for informing targeted, real-time defense mechanisms in robotic platforms.

The findings not only shed light on the vulnerabilities posed by AI-assisted attack methods but also offer a powerful detection strategy that synergizes advanced feature engineering optimized for ROS 2 sensor dynamics, deep neural architectures that effectively leverage multi-modal sensor data, and interpretable analysis techniques that provide actionable security insights. By demonstrating high performance DoS detection performance on the Turtlebot4, this research underscores the potential for robust, real-time security frameworks in next-generation robotic systems.

III. UNMANNED GROUND VEHICLE

This work employs a Turtlebot4 UGV (Fig. 1a), a differential-drive robot equipped with an OAK-D stereo camera, 2D LiDAR, IMU, optical floor tracker, wheel encoders, and Wi-Fi connectivity [23]. The platform runs Ubuntu 22.04 (ROS 2 Humble), publishing sensor data as ROS topics. Its computing architecture comprises a Raspberry Pi 4B and a Create 3 processor, each connected to a separate wireless network, thus enabling remote access and control (Fig. 1b).



(a) Turtlebot4. Source: [23] (b) Connections. Source: [24]

Fig. 1: (a) and (b) depict the overall setup and communication connections diagram of the TurtleBot4.

IV. DoS SIMULATION FRAMEWORK

Recent advancements in LLMs, such as GPT-3 [25], OpenAI ChatGPT [26], and Anthropic Claude, have substantially broadened the potential for adversarial exploits in

robotics [27]. Although these models are generally designed with safety alignment [28], the widespread availability of both uncensored and nominally censored variants still poses considerable security threats. Attackers can leverage even so-called “aligned” models to rapidly generate malicious content [13], bypassing ethical guardrails through carefully crafted prompts [28].

To demonstrate how these vulnerabilities translate into a practical DoS scenario, the following subsections outline the main stages of the proposed attack framework:

- 1) **LLM-Assisted Attack Generation:** To illustrate these risks, the approach in this work employed Anthropic Claude 3 Haiku [22] to generate a multi-threaded Python script aimed at overwhelming ROS 2 communication. By strategically using prompt-engineering methods, it was possible to obtain code that spawns multiple ROS publisher threads. Each thread targets a `/flood/topic_i` channel, configured with aggressive Quality of Service (QoS) parameters (`BEST_EFFORT`, `KEEP_LAST=1`, `VOLATILE`). The script progressively amplifies its traffic by increasing the burst size (up to 30% every few seconds) while reducing inter-burst intervals, emulating a distributed DoS.
- 2) **Attack Execution and Validation:** This script was executed on a separate machine within the Turtlebot4’s ROS 2 network, closely mirroring a genuine external DoS scenario. Observed effects included high CPU usage, increased packet drops, and eventual robot unresponsiveness. Despite nominal safeguards in the LLM, certain prompt formulations partially circumvented built-in restrictions, underscoring persistent vulnerabilities in LLM architectures when subjected to adversarial queries.
- 3) **Implications:** The findings demonstrate how AI-driven DoS attacks can exploit legitimate communication pathways, demanding robust detection and mitigation strategies. Traditional signature-based defenses may become obsolete against rapidly evolving LLM-assisted attacks; therefore, adaptive, context-aware methods—such as the deep learning-based approaches described in this study—are essential for sustaining security in next-generation robotic systems.

V. DATA COLLECTION FRAMEWORK

This research involved collecting data from the Turtlebot4 operating under both normal conditions and during DoS attacks. The data collection sessions took place in a controlled environment, where DoS attacks were systematically executed by publishing high-frequency messages to a set of ROS 2 topics named `/flood/topic_i`. Interactions with the robot were captured using ROS bags, which recorded all messages published by the various ROS nodes throughout each session.

VI. DATA ANALYSIS AND FEATURE DISTRIBUTION

The dataset comprises one normal operation sequence, reflecting standard robotic behaviors, and one attack sequence featuring orchestrated DoS attacks of varying intensities. All interactions were recorded in ROS 2 bags, capturing the complete message flow across the robot’s distributed system. In

this setup, the robot utilizes 33 ROS topics for communicating sensor data and control messages. Each bag includes timestamped messages, topic metadata, and raw message content, yielding a total dataset size of 3.8 GB. From this dataset, 49% of the data represents DoS attack behavior, whereas the remaining 51% corresponds to normal operations.

A. Data Cleaning and Processing

All artificially introduced flood topics were removed to enable generalization beyond a single known attack vector. Messages were then binned in 1-second intervals, with missing values imputed as zeros. Finally, a standard scaling procedure (zero mean, unit variance) was applied across all numeric features. This ensured consistent input scales and mitigated issues caused by the large dynamic ranges often found in network traffic.

B. Feature Distribution

After cleaning and feature engineering, the data were analyzed to characterize message rates and topic usage patterns. The resulting dataset captures both general usage (e.g., message ratio) and more nuanced behaviors (e.g., burstiness, topic entropy), providing insight into whether traffic is heavily skewed toward certain topics or exhibits abrupt surges. As shown in subsequent sections, these distributional properties distinguish normal behavior from DoS attack patterns and thus form the foundation for reliable anomaly detection.

C. Message Rate Distribution Analysis

As shown in Table I, the mean feature values of key ROS topics differ notably between normal and attack scenarios. The overall message rate (`total_msgs`) is reduced during attacks, indicating a system-level impact on throughput. Additionally, the ratio-based features for critical topics, such as `odom_ratio`, `cmd_vel_ratio`, `imu_ratio`, `scan_ratio`, and `tf_ratio`, reveal how individual topics fare relative to the total message volume under varying network loads.

In this work, the ratio for a given topic, such as `/odom`, is computed by normalizing its message count within a 1-second bin to the overall message count in the same bin. Mathematically, this can be expressed as (1):

$$\text{odom_ratio} = \frac{\text{message_count}(\text{/odom})}{\text{total_msgs} + \varepsilon}, \quad (1)$$

where ε is a small constant (e.g., 10^{-9}) added to avoid division by zero. This ratio effectively captures how much bandwidth a particular topic consumes relative to the total throughput.

Comparing these ratio metrics across normal and attack scenarios illuminates the degree to which specific subsystems—such as odometry (`/odom`), velocity commands (`/cmd_vel`), or inertial measurements (`/imu`)—retain priority or degrade during DoS conditions. For instance, if a high-value topic like `/scan` experiences a sudden drop in its ratio, it suggests that the robot’s mapping or obstacle avoidance capacity could be severely compromised. By contrast, topics that show an increased ratio during an attack may be absorbing a disproportionate share of available resources, further exacerbating performance bottlenecks in other components.

TABLE I: Mean Feature Values by Scenario

	total_msgs	odom_ratio	cmd_vel_ratio	imu_ratio	scan_ratio	tf_ratio
attack	401.3642	0.0375	0.0647	0.0621	0.0039	0.0063
normal	528.8404	0.0331	0.0600	0.0563	0.0150	0.0148

Taken together, these findings underscore the importance of monitoring message distribution trends as part of a robust cybersecurity strategy for robotics.

D. Feature Engineering

The feature engineering phase focused on capturing both general and domain-specific patterns in network traffic. A standard scaling procedure was applied to all numeric variables (e.g., `total_msgs`, `odom_ratio`, `imu_ratio`) to normalize them, ensuring that no single feature overshadows the others due to large value ranges. This normalization not only stabilizes model training but also facilitates interpretability—especially when applying integrated gradients—by making feature contributions more comparable.

Two key features were introduced to move beyond basic message counts and better detect anomalous patterns:

- **Burstiness (B):** As defined in (2), it is the difference between the maximum and average number of messages in each one-second bin,

$$B = \text{max_count} - \text{mean_count}, \quad (2)$$

which highlights abrupt spikes in traffic, a signature of DoS or DDoS attacks. Even if total message volume appears normal, a sudden surge over a short interval may overwhelm system resources, making burstiness an effective indicator of transient overloads.

- **Topic Entropy (H):** Computed using Shannon’s entropy formula [29], shown in (3):

$$H = - \sum_{i=1}^N p_i \ln(p_i), \quad (3)$$

where p_i is the fraction of messages within a bin attributed to topic i . This measure captures how evenly traffic is distributed across various channels. A high-entropy state indicates diverse, well-balanced communication, whereas a low-entropy state suggests that only one or two topics dominate. Sudden drops in entropy may signal malicious redirection of traffic or the monopolization of bandwidth by a targeted flood.

Artificially generated `/flood/topic_i` data were excluded from the final feature set to avoid overfitting to a single known DoS channel and to promote broader model generalization. Real-world adversaries may exploit any legitimate ROS 2 topic, rather than one explicitly labeled as `/flood/topic_i`. By omitting these synthetic channels, the feature set better reflects diverse attack vectors and increases resilience against DoS attacks initiated through alternative network paths.

Collectively, these engineered features and the careful exclusion of `/flood` traffic enable monitoring of key dynamics such as sudden bursts or abnormal topic distributions. As

KDE Distributions: Burstiness & Topic Entropy (Normal vs. Attack)

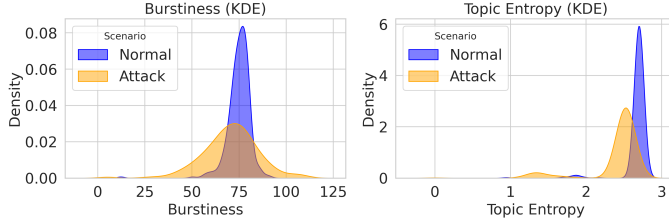


Fig. 2: *Kernel density estimation* (KDE) plots showing the distributions of the burstiness (i.e., `max_count - mean_count` in message counts) and `topic_entropy` (dimensionless Shannon measure of the distribution of messages across active topics) under normal and attack scenarios. The differing peaks and spread of these curves highlight how these two features can help discriminate between benign and malicious behavior.

illustrated in Fig. 2, the kernel density estimation (KDE) plots of `burstiness` and `topic_entropy` exhibit distinct patterns for normal versus attack scenarios, thereby supporting timely detection and response.

E. Training Split and Final Feature Set

After the data cleaning and feature engineering steps were completed, the dataset was divided into training and testing partitions for model evaluation. The dataset was split into training (60%), validation (20%), and test (20%) sets, maintaining the class distribution across all partitions. This approach preserves proportional representation of normal and attack instances, helping to maintain balanced conditions for supervised learning.

a) *Feature Selection.*: All feature columns described in Section VI-D are retained, including:

- **Message-Based Features:** `total_msgs`, `burstiness`, and `topic_entropy`, reflecting overall traffic intensity and distribution.
- **Ratio-Based Features:** `odom_ratio`, `cmd_vel_ratio`, `imu_ratio`, `scan_ratio`, and `tf_ratio`, each quantifying how much bandwidth a specific operational topic consumes relative to total traffic in a 1-second bin.

b) *VAE Training.*: For the unsupervised VAE, only normal data from the training portion is used to learn a latent representation of benign traffic patterns. Attack samples are later assessed based on a reconstruction error threshold—determined from the distribution of reconstruction errors on the normal training set (95th percentile). Any test sample whose error exceeds this baseline is classified as anomalous. By decoupling itself from explicit labels during training, the VAE remains flexible for detecting novel or unexpected attack behaviors in the test set. See Section VII for further architectural details.

The outcome of these steps is a *tabular dataset* comprising numeric columns and, for supervised models, a binary class

label (normal or attack). This final feature set serves as the input for the neural architectures discussed in Section VII.

VII. MODEL ARCHITECTURE

Table II summarizes the four neural architectures employed for DoS attack detection and outlines their respective hyperparameters. These models were chosen to cover a spectrum of design philosophies—ranging from a straightforward fully connected approach (MLP) to more sophisticated sequence-based networks (LSTM, 1D CNN) and an unsupervised latent-space model (VAE):

- 1) **MLP**: A neural architecture that operates on per-instance features. The MLP provides a direct comparison for how well feed-forward networks handle tabular input (e.g., `total_msgs`, `burstiness`, `topic_entropy`). While not explicitly modeling temporal dependencies, it frequently achieves competitive performance on structured data and offers a simpler path for interpretability using IG.
- 2) **LSTM**: This recurrent model addresses the temporal dimension by processing short sequences (5s in this work). Its internal gating mechanisms are well-suited for capturing temporal patterns, such as short bursts of malicious traffic or gradual drifts in normal activity. These properties are expected to enhance the detection of time-dependent DoS signatures.
- 3) **CNN (1D)**: A one-dimensional time series convolutional network is employed to process fixed-length sequences of feature values, treating time as the single spatial dimension. This design is particularly suitable for time-series data in which each time step contains a vector of features (e.g., message counts, ratios, entropy). Compared to an LSTM, a 1D CNN typically trains faster and excels at capturing localized spikes or abrupt transitions in traffic, though it may not preserve long-range dependencies as effectively. In contrast to the earlier studies in [30] and [31], where the authors transformed the data into images as an attack signature, this research directly utilizes time series data, preserving its inherent temporal structure and avoiding any potential information loss that may arise from image transformations. In the context of DoS detection, the 1D CNN can highlight short-duration bursts or distinctive local patterns in the feature time series.
- 4) **VAE**: An unsupervised approach that learns a compact latent representation of normal traffic patterns through reconstruction. In many real-world scenarios, labeled DoS data can be sparse or entirely unavailable, making an unsupervised strategy especially advantageous. By measuring the reconstruction error—how closely the VAE can reproduce its input—the model flags anomalies when inputs substantially deviate from the learned “normal” latent distribution. This characteristic is particularly valuable for identifying novel or unexpected attack vectors that may not appear in the training set.

These architectural choices facilitate coverage of both high-level statistical features (MLP, VAE) and temporal dynamics (LSTM, CNN). In addition, each model supports integrated

gradients analysis, enabling a deeper examination of how input features or time steps contribute to classification or reconstruction. The following section (VIII) reports the empirical performance of these models, including quantitative evaluation metrics, and then proceeds with a dedicated subsequence on model interpretability using integrated gradients.

TABLE II: Summary of Model Architectures¹

Model	Architecture
MLP	Layers: Input($input_dim$) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Output: Dense(1, Sigmoid)
LSTM	Layers: Input($seq_len=5$, $input_dim$) $\rightarrow$ LSTM(64, dropout=0.2) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Output: Dense(1, Sigmoid)
CNN (1D)	Layers: Input($seq_len=5$, $input_dim$) $\rightarrow$ Conv1D(32, kernel=2, ReLU) $\rightarrow$ MaxPool1D(2) $\rightarrow$ Flatten $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Output: Dense(1, Sigmoid)
VAE	Encoder: Dense(32, ReLU) $\rightarrow$ (z_{mean} , z_{log_var}) $\rightarrow$ Reparameterize Decoder: Dense(32, ReLU) $\rightarrow$ Output: Dense($input_dim$) Latent Dim: 8

¹ All models were trained for 50 epochs with a batch size of 16 and the Adam optimizer, using early stopping (patience=3) and checkpointing (ModelCheckpoint).

VIII. RESULTS

A. Quantitative Results

Table III and Fig. 3 compare the accuracy, precision, recall, and F1-score of the four candidate models (MLP, LSTM, CNN, VAE) on the held-out test data. Fig. 4 further illustrates ROC curves, highlighting each model’s AUC. The following subsections discuss major findings for each model:

TABLE III: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1	AUC
MLP	0.97	0.97	0.96	0.97	0.998
LSTM	0.91	0.93	0.88	0.90	0.985
CNN	0.61	0.57	0.70	0.63	0.625
VAE	0.64	0.86	0.33	0.48	0.863

a) MLP: The MLP substantially outperforms the other architectures in both overall accuracy (97%) and balanced metrics such as precision and recall, each above 0.97. Its confusion matrix (Table III) shows minimal misclassification across normal and attack classes. Two factors likely contribute to this strong performance: (1) the MLP effectively leverages the engineered features (*burstiness*, *topic_entropy*, etc.) in a tabular setting, and (2) the dataset may not contain sufficient temporal nuances to favor sequence models (LSTM/CNN). Despite lacking explicit temporal modeling, the MLP’s ability to capture static feature interactions appears sufficient to detect DoS attacks in this dataset.

b) LSTM: The LSTM model attains a robust accuracy of 91%, with a high F1-score of 0.90 and an AUC of 0.985. These results highlight the effective temporal modeling that LSTM’s gating mechanisms can provide, even within relatively short 5-second windows. The model’s recall of 0.88 and precision of 0.93 indicate that it successfully identifies the majority of

“Attack” instances while keeping false positives comparatively low. This balance underscores the LSTM’s strong capacity for capturing essential sequence dependencies and differentiating normal from malicious behavior in a DoS detection setting.

c) CNN: The 1D CNN remains below the performance of both the MLP and LSTM, achieving an accuracy of 61% with an F1-score of 0.63 and an AUC of 0.625. While the CNN attains a moderate recall of 0.70 for “Attack”, its precision of 0.57 indicates a tendency toward false positives. This gap underscores that, despite the CNN’s aptitude for extracting local temporal patterns, constraints on sequence length and limited training diversity likely hinder its ability to fully capture subtle or short-lived anomalies in a DoS setting. More extensive data, deeper architectures, or longer sequences may be required to realize the full potential of CNN-based models in this domain.

d) VAE: The VAE operates under an unsupervised paradigm, focusing on reconstruction error as an anomaly indicator. Despite its moderate overall accuracy of 64% and an F1-score of 0.48, the model exhibits high precision (0.86) but low recall (0.33) for attack detection. In other words, while it rarely misclassifies normal traffic as attacks, it fails to detect many actual attack instances that do not manifest as large reconstruction errors. This outcome suggests that although the VAE captures standard traffic patterns well, certain DoS anomalies fall below its anomaly threshold. Nevertheless, the AUC of 0.863 indicates that with appropriate threshold tuning, the VAE could more flexibly balance its precision–recall trade-off to meet specific operational demands.

e) Ablation: To further justify the impact of each feature, this research performed a small ablation study in which one or more features (e.g., *burstiness*, *topic_entropy*) were removed and the models were retrained. In particular, removing *topic_entropy* from the MLP feature set reduced its overall accuracy from 0.97 to 0.92, while removing both *topic_entropy* and *burstiness* lowered it to 0.89. These drops indicate that the engineered metrics, which capture sudden spikes or skewed topic usage, significantly contribute to distinguishing DoS attacks from normal traffic. Conversely, the LSTM and CNN were less sensitive to the removal of these features, suggesting that their performance gains are more influenced by hyperparameter choices (e.g., sequence length, kernel size) rather than individual feature importance.

f) Overall Observations: Among the four tested architectures, the MLP not only achieves the highest and most balanced performance (97% accuracy, with strong precision, recall, and F1 metrics) but also detects attacks more quickly since it does not require the 5-second temporal window needed by the sequence models. The LSTM likewise performs well at 91% accuracy, demonstrating its ability to leverage temporal cues even within relatively short sequences, although with slightly delayed detection due to its sequence-based nature. By contrast, the CNN yields 61% accuracy, suggesting that local convolutional filters may be insufficient for capturing the breadth of DoS anomalies. Finally, the VAE operates with 64% accuracy, underscoring the difficulty of identifying attacks solely through reconstruction error—though its AUC indicates

potential for improved detection with threshold tuning. Together, these outcomes highlight the critical role of data representation, sequence framing, and alignment between model capacity and the temporal characteristics of DoS attacks.

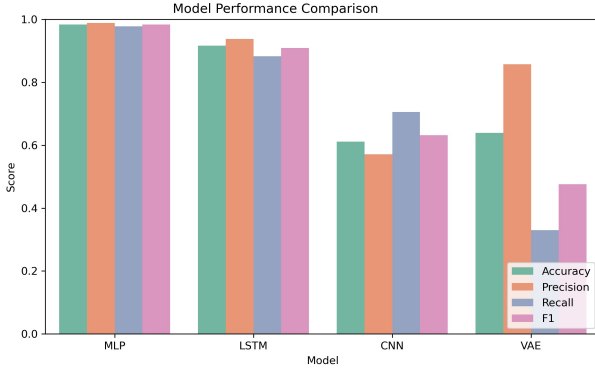


Fig. 3: Bar chart of accuracy, precision, recall, and F1-score for all four models

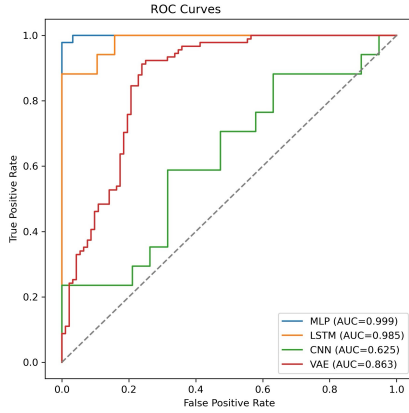


Fig. 4: ROC curves comparing MLP, LSTM, CNN, and VAE. The MLP demonstrates high performance in detection ($AUC \approx 0.999$), whereas CNN yields the lowest AUC (0.625).

B. Model Interpretability

To better understand how each architecture arrives at its classification or reconstruction decisions, this work applies *Integrated Gradients* (IG) analysis. IG quantifies each input feature's contribution to the final output (e.g., predicted probability or reconstruction error) by integrating gradients between a chosen baseline (zero) and the original input. Although IG analysis was carried out on multiple test samples, only one representative example per model is shown (Figs. 5–8) for brevity. The observed trends generalize well across other samples, consistently highlighting *odom_ratio*, *scan_ratio*, and *imu_ratio* as influential features.

a) MLP (Fig. 5): The bar chart reveals that *odom_ratio* and *scan_ratio* have the greatest positive impact on the MLP's output, while *topic_entropy* exerts a negative contribution. This suggests that, for the given sample, the MLP relies heavily on spatial or distance-related

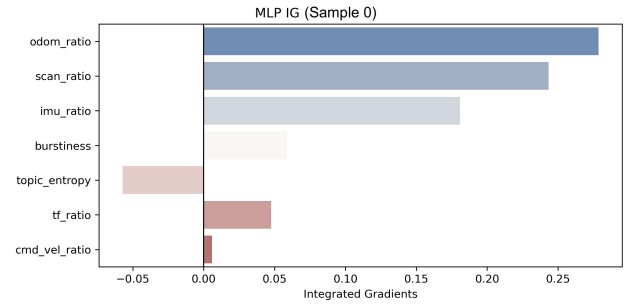


Fig. 5: MLP IG Bar Chart (Sample 0). Higher positive values indicate that *odom_ratio* and *scan_ratio* most strongly increase the probability of predicting a specific class. Negative values, such as *cmd_vel_ratio*, have an opposing influence.

features to identify potential DoS anomalies. By contrast, *cmd_vel_ratio* appears to reduce the model's propensity for an "attack" classification.

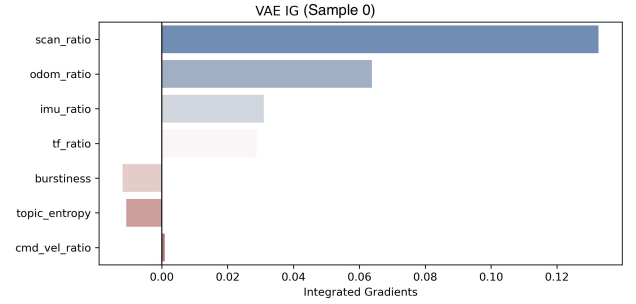


Fig. 6: VAE IG Bar Chart (Sample 0). Here, *scan_ratio* and *odom_ratio* dominate the reconstruction error, indicating sensitivity to sensor-specific data.

b) VAE (Fig. 6): Although it does not rely on labels during training, the VAE still provides an interpretability signal via reconstruction error gradients. As shown, *scan_ratio* and *odom_ratio* drive most of the error for this sample, suggesting these sensor streams deviate most from the VAE's learned "normal" distribution. Lower contributions from *topic_entropy* and *burstiness* imply that, in this instance, the overall message distribution is less relevant to the anomaly signal.

c) LSTM (Fig. 7): The integrated gradients heatmap for the LSTM shows that much of its decision hinges on later time steps ($t3-t4$), with strong positive attributions for *odom_ratio* and *scan_ratio*. This suggests the LSTM responds to changes in robot motion and sensor data near the end of the 5-second window—potentially indicative of a developing DoS condition. Overall, the LSTM's sequence-based design allows it to highlight these late-stage signals, indicating how temporal context can be pivotal for distinguishing normal from suspicious message patterns.

d) CNN (Fig. 8): The integrated gradients heatmap indicates that the CNN relies on short, localized patterns in its sequence data. In particular, a strong positive attribution appears

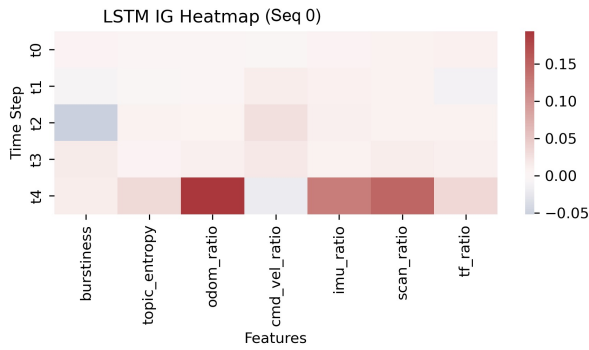


Fig. 7: LSTM IG Heatmap (Seq 0). Each cell shows the gradient-based importance of a particular feature (columns) at a specific time step (rows). Warm tones indicate positive influence on the output; cool tones indicate negative influence.

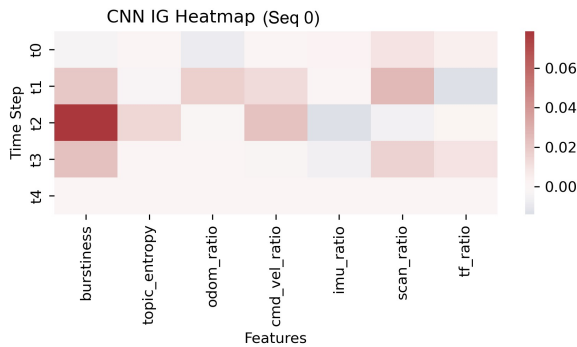


Fig. 8: CNN IG Heatmap (Seq 0). The 1D CNN extracts local patterns over short windows. Red indicates features/time steps that increase the likelihood of “attack,” whereas blue indicates features/time steps that decrease it.

at t_2 for *burstiness*, suggesting that sudden fluctuations in message distribution at that moment play a key role in triggering an “attack” prediction. Conversely, blue-shaded cells (negative attributions) reveal feature–time combinations that lower the model’s confidence, such as certain *odom_ratio* values around t_2 – t_3 . Overall, this pattern of localized sensitivity aligns with the CNN’s reliance on convolution filters, which naturally attend to brief but salient signals rather than extended temporal dependencies.

e) Summary of Interpretability Findings: Across all architectures, *odom_ratio*, *scan_ratio*, and *imu_ratio* consistently emerge as top drivers of classification or reconstruction decisions, reflecting their importance in detecting irregularities in robot motion and sensor streams. By contrast, *burstiness* and *topic_entropy* exhibit more varied contributions, appearing crucial in certain samples but less so in others. The sequence-based models (LSTM, CNN) tend to focus on short-term changes in these sensor features—often at specific time steps—while the MLP and VAE react more to broader or static-level patterns. These interpretability insights support the quantitative evaluations in Section VIII-A, showing how different architectures leverage feature–time interactions

to discriminate normal from attack traffic.

IX. CONCLUSION

The research presented in this paper proposes a deep learning-based framework for detecting DoS attacks on a ROS 2-enabled UGV robot. By leveraging LLM-assisted adversarial script generation, domain-specific feature engineering, and controlled data collection, the results show that a feed-forward MLP can attain high performance (97% accuracy, $AUC \approx 0.999$) in classifying DoS traffic. Notably, the LSTM also performs at a high level (91% accuracy, $AUC 0.985$), demonstrating the value of temporal modeling when the data distribution permits. By contrast, the CNN (61% accuracy, $AUC 0.625$) appears more sensitive to short windows and insufficient training examples, while the unsupervised VAE (64% accuracy, $AUC 0.863$) underscores the challenge of relying on reconstruction error alone for detecting subtle or short-lived attacks. Overall, the findings suggest that—beyond network depth or sequence modeling—well-crafted metrics (e.g., *burstiness*, *topic_entropy*) are pivotal for capturing sudden traffic surges and imbalances across ROS 2 topics, leading to more reliable DoS detection in real-world robotic scenarios.

The results demonstrate that this approach achieves near real-time DoS detection in physical ROS 2-enabled UGV deployments, maintaining both high accuracy (97%) and interpretability through Integrated Gradients analysis. By leveraging ROS 2-specific traffic patterns with a one-second binning strategy, the system can quickly identify attacks through sudden traffic surges (captured by *burstiness*) and imbalances across ROS 2 topics (measured by *topic_entropy*), particularly in critical sensor streams like *odom* and *scan*. This rapid detection capability, combined with high accuracy and low latency, makes the system suitable for safety-critical robotic operations where even brief disruptions could have severe consequences.

Beyond this specific case, the proposed pipeline offers a scalable blueprint for enhancing cybersecurity in networked robotic and IoT systems. By integrating interpretability methods like Integrated Gradients, stakeholders can gain actionable insights into which sensor streams or traffic patterns signal malicious behavior, enabling swift mitigation strategies in safety-critical domains such as healthcare, autonomous driving, and industrial automation. Future work will extend these methods to diverse robotic platforms (e.g., multi-UGV or aerial swarms) and more complex attack vectors, including low-rate or multi-vector intrusions across multiple network layers. Additionally, deploying the detection and interpretability modules on resource-constrained hardware is essential for real-world readiness. Overall, the synergy of domain-specific features, deep learning architectures, and explainable AI provides a promising path toward robust, scalable defense mechanisms for next-generation autonomous systems.

REFERENCES

- [1] A. Khazetdinov, A. Aleksandrov, A. Zakiev, E. Magid, and K. Hsia, “RFID-based warehouse management system prototyping using a heterogeneous team of robots,” *Robots in human life*, vol. 263, pp. 2020–24, 2020.

- [2] J. Szrek, R. Zimroz, J. Wodecki, A. Michalak, M. Góralczyk, and M. Worsa-Kozak, "Application of the infrared thermography and unmanned ground vehicle for rescue action support in underground mine—the amicos project," *Remote Sensing*, vol. 13, no. 1, p. 69, 2020.
- [3] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Science robotics*, vol. 7, no. 66, eabm6074, 2022.
- [4] B. Dieber, B. Breiling, S. Taurer, S. Kacianka, S. Rass, and P. Schartner, "Security for the robot operating system," *Robotics and Autonomous Systems*, vol. 98, pp. 192–203, 2017.
- [5] J. Cannady, "Artificial neural networks for misuse detection," in *Proceedings of the 1998 National Information Systems Security Conference (NISSC'98)*, 1998, pp. 443–456.
- [6] J. Kang, H. Yang, Y. Zhang, Y. Dai, M. Zhan, and W. Wang, "Actdetector: A sequence-based framework for network attack activity detection," in *2022 IEEE Symposium on Computers and Communications (ISCC)*, IEEE, 2022, pp. 1–7.
- [7] G. Welderman, R. Castellanos, A. Whitacre, F. Montague, and J. Starck, "A robust system for ransomware detection using temporal behavior modeling," 2024.
- [8] F. J. Rendón-Segador, J. A. Álvarez-García, and A. J. Varela-Vaca, "Paying attention to cyber-attacks: A multi-layer perceptron with self-attention mechanism," *Computers & Security*, vol. 132, p. 103 318, 2023.
- [9] T. Bilot, N. El Madhoun, K. Al Agha, and A. Zouaoui, "Graph neural networks for intrusion detection: A survey," *IEEE Access*, vol. 11, pp. 49 114–49 139, 2023.
- [10] T. Ji, A. N. Sivakumar, G. Chowdhary, and K. Driggs-Campbell, "Proactive anomaly detection for robot navigation with multi-sensor fusion," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4975–4982, 2022.
- [11] S. Khan, C. F. Liew, T. Yairi, and R. McWilliam, "Unsupervised anomaly detection in unmanned aerial vehicles," *Applied Soft Computing*, vol. 83, p. 105 650, 2019.
- [12] Z. Pethő, T. M. Kazár, Z. Szalay, and Á. Török, "Quantifying cyber risks: The impact of dos attacks on vehicle safety in V2X networks," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [13] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (llm) security and privacy: The good, the bad, and the ugly," *High-Confidence Computing*, p. 100 211, 2024.
- [14] M.-C. Popescu, V. E. Balas, L. Perescu-Popescu, and N. Mastorakis, "Multilayer perceptron and neural networks," *WSEAS Transactions on Circuits and Systems*, vol. 8, no. 7, pp. 579–588, 2009.
- [15] A. Graves and A. Graves, "Long short-term memory," *Supervised sequence labelling with recurrent neural networks*, pp. 37–45, 2012.
- [16] K. O'Shea, "An introduction to convolutional neural networks," *arXiv preprint arXiv:1511.08458*, 2015.
- [17] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," *Special lecture on IE*, vol. 2, no. 1, pp. 1–18, 2015.
- [18] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International conference on machine learning*, PMLR, 2017, pp. 3319–3328.
- [19] A. Mustapha *et al.*, "Detecting ddos attacks using adversarial neural network," *Computers & Security*, vol. 127, p. 103 117, 2023.
- [20] M. Kordestani and M. Saif, "Observer-based attack detection and mitigation for cyberphysical systems: A review," *IEEE Systems, Man, and Cybernetics Magazine*, vol. 7, no. 2, pp. 35–60, 2021.
- [21] R. Ma, X. Chen, and R. Zhai, "A DDoS attack detection method based on natural selection of features and models," *Electronics*, vol. 12, no. 4, p. 1059, 2023.
- [22] Anthropic, *Models overview*, Technical Documentation, Accessed: 2024-11-12, Anthropic, 2024. [Online]. Available: <https://docs.anthropic.com/en/docs/models-overview>.
- [23] Clearpath Robotics, *Turtlebot 4*, Accessed: 2024-12-09, Clearpath Robotics, 2024. [Online]. Available: <https://clearpathrobotics.com/turtlebot-4/>.
- [24] Clearpath Robotics, *Overview*, Accessed: 2024-12-16, Clearpath Robotics, 2024. [Online]. Available: <https://turtlebot.github.io/turtlebot4-user-manual/software/overview.html>.
- [25] T. B. Brown, "Language models are few-shot learners," *arXiv preprint ArXiv:2005.14165*, 2020.
- [26] K. I. Roumeliotis and N. D. Tselikas, "Chatgpt and open-ai models: A preliminary review," *Future Internet*, vol. 15, no. 6, p. 192, 2023.
- [27] T. Orchard and L. Tasiemski, "The rise of generative ai and possible effects on the economy," *Economics and Business Review*, vol. 9, no. 2, pp. 9–26, 2023.
- [28] W. Lu *et al.*, "Eraser: Jailbreaking defense in large language models via unlearning harmful knowledge," *arXiv preprint arXiv:2404.05880*, 2024.
- [29] S. Vajapeyam, "Understanding shannon's entropy metric for information," *arXiv preprint arXiv:1405.2061*, 2014.
- [30] F. Santoso and A. Finn, "A data-driven cyber-physical system using deep-learning convolutional neural networks: Study on false-data injection attacks in an unmanned ground vehicle under fault-tolerant conditions," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 1, pp. 346–356, 2023.
- [31] F. Santoso and A. Finn, "Trusted operations of a military ground robot in the face of man-in-the-middle cyberattacks using deep learning convolutional neural networks: Real-time experimental outcomes," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 4, pp. 2273–2284, 2024.